| Document No. | QSL-CAL-PROC-003 |
|---|---|
| Revision / Date | 1.0 — 26 June 2026 |
| Approved By | Technical Manager |

Qalibrated Systems Limited

# CALIBRATION PROCEDURE

Temperature Indicating & Sensing Devices — per EURAMET Calibration Guide No. 13

## 1. Scope & Application

This procedure applies to the calibration of temperature indicating and sensing devices — including digital thermometers, thermocouples, and resistance thermometers used in calibration, inspection, and process monitoring — over the range −30°C to +300°C as typically encountered in QSL's scope of work. It is based on the principles of EURAMET Calibration Guide No. 13, "Calibration of Temperature Block Calibrators", adapted to comparison calibration of sensors and indicators in a temperature bath or block calibrator.

## 2. Reference Standards & Traceability

| STANDARD | DESCRIPTION | TRACEABILITY |
|---|---|---|
| Reference Platinum Resistance Thermometer (PRT) | Calibrated reference probe spanning the required test range | National metrology institute, ≲ 24 months |
| Temperature Block Calibrator / Bath | Stable temperature source with adequate well depth for both reference and test probes | Characterised for stability and uniformity |
| Precision Indicator / Bridge | Readout for the reference PRT | Accredited lab, ≲ 24 months |

## 3. Environmental Conditions

| PARAMETER | REQUIREMENT |
|---|---|
| Stability | Source temperature stable to within the required tolerance before any reading is recorded |
| Immersion | Reference and test probes immersed to the same depth, as close together as practicable without contact |
| Ambient Conditions | Room temperature recorded; draughts avoided around the calibration source |

## 4. Procedure

1. **Pre-Test Inspection.** Inspect the instrument under test for physical damage, secure connections, and correct display function across its range.
2. **Test Point Selection.** Select a minimum of 3–5 test temperatures distributed across the instrument's calibration range, including points near the extremes of typical use.
   *Per cg-13 principle: investigations carried out at the operating temperature showing the greatest difference from ambient (both positive and negative) capture the most significant influences.*
3. **Stabilisation.** At each test point, allow the source and both probes to reach thermal equilibrium, confirmed by a stable reference reading over a defined dwell period.
4. **Comparison Reading.** Record the reference PRT reading and the indication of the instrument under test simultaneously (or in immediate succession) at each test point, repeating a minimum of 3 times per point.

5. **Uniformity Check (where applicable).** Where the source is a block calibrator or bath with multiple wells, assess temperature uniformity between wells used for the reference and test probes.
6. **Calculate Error.** Determine the error of the instrument under test as the mean indicated value minus the corrected reference value at each test point.

## 5. Uncertainty & Calculation

The combined standard uncertainty includes: the calibration uncertainty of the reference PRT; the resolution of the instrument under test; the repeatability of readings at each test point; the source stability and uniformity (inhomogeneity between probe positions); and any self-heating or immersion effects. These are combined per the GUM method to give the expanded uncertainty (k=2) of the error at each test point.

## 6. Acceptance Criteria

The instrument passes calibration where the error at each test point, together with its uncertainty, falls within the manufacturer's specification or the client's stated tolerance. Instruments exceeding tolerance at any test point are reported as out of tolerance; where the instrument supports user adjustment, this is noted separately from the as-found calibration result.

| Prepared By | Reviewed By | Approved By |
|---|---|---|
| | | |
| Technical Manager | Quality Manager | Managing Director |